

Day 7 — Sāyana vs Nirayana

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: distinguish tropical and sidereal zodiacs; introduce ayanāṁśa.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *nirayana* (IAST: nirayana; lit. “sidereal zodiac (linked to fixed stars)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Set up the conflict.** Indian astronomy distinguishes two reference frames:
 - *Sāyana* (with-ayana, “tropical”) — measured from the Vernal Equinox.
 - *Nirayana* (without-ayana, “sidereal”) — measured from a fixed point in the sky (a star, typically near Citrā/Spica).
2. **The drift.** Earth’s axis wobbles (precession of equinoxes), causing the Vernal Equinox point to shift slowly westward against the fixed stars at ~50 arcseconds per year. Over ~2000 years, the cumulative drift is ~28°. *That* is the ~24° gap between Dec 21 (solstice in sāyana) and Jan 14 (when the Sun enters Makara in nirayana).
3. **Source.** Āryabhaṭīya documents the precession concept; Bhāskara II refines it. Indian jyotiṣa preserves the *nirayana* convention; modern astronomy uses *sāyana*.
4. **Why does this matter?** Festival dates anchored to nirayana drift slowly relative to the seasons. Over thousands of years, *Makara Saṅkrānti* will shift even further from the December solstice unless re-anchored.
5. **Critical thinking.** “*Should jyotiṣa update its zodiac to sāyana? Or preserve the nirayana convention for cultural continuity?*” Both views have advocates. There’s no single right answer.

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Sāyana vs Nirayana.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).